A Two-Stage Mutual Fusion Network for Multispectral and Panchromatic Image Classification

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***Abstract*— With the rapid development of remote sensing technology, satellites can easily obtain multispectral (MS) and panchromatic (PAN) images. How to mine the essence and peculiarity of the MS and PAN images and utilize their complementary to improve the classification performance is still a challenge. This article designs a two-stage mutual fusion network (TSMF-Net) for MS and PAN image classification. The network can be divided into two stages: data fusion and feature fusion. In the data fusion stage, we propose an adaptive twin intensity–hue–saturation (ATIHS) strategy. It not only aligns the size and channels of the MS and PAN images by a novel q-Split operation but also introduces an adaptive soft-average mask to reduce the differences between replacement components, effectively mitigating spectral distortion and paving the way for the next stage. In the feature fusion stage, we propose a feature graft block (FG-Block) in which we introduce triplet loss and design an interlaced channel addition (ICA) module. Under the supervision of triplet loss, the FG-Block separates and hauls each branch's essential and peculiar features. With the help of the ICA module, it can effectively graft the essential feature between branches and retain the peculiar feature of each branch, improving the utilization and discrimination of features. Finally, composed of the ATIHS, FG-Blocks, and output layers, our TSMF-Net is proven to improve the accuracy of the remote sensing classification task. The experimental results on multiple datasets verify the effectiveness of our proposed algorithms. Our code is available at: https://github.com/liaoyinuo/TSMF-Net.**

***Index Terms*— Deep learning (DL), deep neural network, feature fusion, multispectral (MS) images, panchromatic (PAN) images, remote sensing.**

I. Introduction

With the support of increasingly advanced remote sensing technology, satellites can obtain increasingly high-quality remote sensing images, especially multispectral (MS) images with rich spectral information and panchromatic (PAN) images with high spatial resolution, such as Pleiades-1A (2-m MS and 0.5-m PAN images), Deimos-2 (4-m MS and 1-m PAN images), KOMPSAT-3A (2.2-m MS and 0.55-m PAN images), WorldView-4 (1.24-m MS and 0.31-m PAN images), and Gaofen-2 (3.2-m MS and 0.8-m PAN images) [1], [2]. However, it is hard to obtain comprehensive scene interpretation and good classification results by analyzing MS or PAN images in isolation. Therefore, how to utilize the complementary between the MS and PAN images to improve the classification performance has become a research hotspot in recent years.

Most multiresolution classification algorithms for PAN and MS images can be divided into two categories: traditional methods and methods based on deep learning (DL).

One of the traditional methods is data fusion first and then classification. It can be classified into four types: component substitution (CS)-based methods [3], multiresolution analysis (MRA)-based methods [4], sparse representation (SR)-based methods [5], and hybrid methods that combine multiple algorithms [6]. The CS methods include the Brovey transform (BT) [7], [8], principal component analysis (PCA) transform [9], [10], and intensity–hue–saturation (IHS) transform [11], [12], [13]. As a typical approach of the CS method, the IHS decomposes the MS image into spatial and spectral components and replaces MS's spatial component with the PAN image to obtain the sharpened MS image. The MRA-based methods include the discrete wavelet transform (DWT) [14], [15], [16], Laplace pyramid transform (LPT) [17], [18], and so on. This kind of method first obtains the feature representation in the joint spatial–spectral domain by multiscale transform, then computes the multiscale coefficients to perform a weighted fusion, and finally performs the inverse multiscale transform to obtain the final results. The SR-based methods [19], [20], [21] cut the image into multiple overlapping patches to maintain translation invariance, reconstructing the fused image through the fusion coefficients and the dictionary. Moreover, there are also some studies combining multiple algorithms for image fusion [22], [23], [24], [25], [26]. Li and Yang [27] propose a hybrid multiresolution approach that combines the wavelets and contourlets for fusion. Liu et al. [28] proposed a framework that decomposes images by the multiscale decomposition method, fuses low-frequency components by SR, and finally performs inverse transformation to obtain the results. The above algorithms are known for their ability to obtain feasible fusion results quickly. However, they tend to lose detailed

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information and may even suffer from serious distortion problems in some cases.

Another traditional method is feature fusion first and then classification. Storvik et al. [29] proposed a Bayesian classification framework based on a multiscale model and realized the classification by an iterative conditional mode algorithm. Voisin et al. [30] proposed a contextual approach based on the Markov random field, using linear mixed models and graph cutting methods to generate contextual classification graphs. Huang et al. [31] extracted the feature by the CLBP descriptors and Fisher vector coding and introduced a kernel-based extreme learning machine for classification. Tuia et al. [32] considered the spectral and spatial information, using a support vector machine with a composite kernel to extract features. These methods can achieve good results, but the extraction is usually shallow and hard to be comprehensive.

Recently, with the rapid development of the deep neural network, methods based on DL are drawing more and more attention [33], [34], [35], [36], [37]. Zhang et al. [38] proposed a framework based on multiscale convolutional sparse decomposition (MCSD), reconstructing the fused image by the summation of fused low-frequency and feature maps. Zhu et al. [35] combined the DWT and GIHS methods, proposing a novel three-branch network for classification. Ma et al. [39] first fused remote sensing images by an improved traditional method. Then, they performed classification through a neural network composed of correlation-based attentional feature fusion (CAFF) modules, which is based on the second-order correlation of features. Xing et al. [40] proposed a dual collaborative fusion model, and they designed a novel graph convolution network to modulate interspectral relationships, combining the adjusted MS features with the high-frequency components of the PAN features to obtain the results.

The above studies make courageous explorations and outstanding contributions, achieving encouraging results that inspire our work. However, there are still some easily ignored issues worth attention.

1) There are structural and representational differences between the MS and PAN images. On the one hand, the MS and PAN images inherently have structural differences due to the different spatial and spectral resolutions. The different sizes and channel numbers bring challenges to the design of image fusion algorithms and classification networks. On the other hand, there are representational differences between the MS and PAN images due to their different emphasis on the representation of spatial and spectral domains, and they are easily influenced by atmospheric and illumination factors during the acquisition process. The representational differences widen the gap and increase the network burden. However, the existing algorithms pay little attention to structural differences, and there is still some room for improvement in reducing the representational differences.
2) The MS and PAN images have the same essence of representing the identical region, and they own the respective peculiarity in spatial and spectral domains. Many studies focus on how to extract features from the entirety, but they hardly consider the correlation between essence and the differences between peculiarities. On the one hand, within the branch, it leads to confusion about essential and peculiar features, resulting in information loss. On the other hand, between the branches, the essential features tend to be projected into different feature spaces, causing information redundancy. Therefore, how to separate, haul, and fuse essential and peculiar features is the key to improving the network performance.

Based on the above problems, we propose a two-stage mutual fusion network (TSMF-Net) for MS and PAN image classification. The contributions of this article are given as follows.

1) In order to effectively reduce the structural and representational differences, we propose an adaptive twin IHS (ATIHS) data fusion strategy. For the structural differences, we propose a q-Split operation and a channel alignment strategy in the ATIHS strategy. It makes the MS and PAN images in the same shape and enhances the similarity between their channels, reducing the structural differences effectively. For the representational differences, we propose an adaptive soft-average mask to weight the intensity components of MS and PAN, obtaining a substitution intensity component that is full of information on both sides. The combination of this intensity component and improved edge detection operators for fusion can effectively reduce the representational differences between the MS and PAN images. What is more, our algorithm yields fused MS and PAN images of the same shape, able to adapt more variety of neural networks.
2) We novelly propose a feature graft block (FG-Block) in which we introduce the triplet loss and design an interlaced channel addition (ICA) module. We first introduce the triplet loss for feature mining, aiming to separate and haul the essence and peculiarity of features. After that, we need to fuse each branch's essential and peculiar features to further strengthen and integrate them. Therefore, we propose the ICA module, which can make full use of the correlation between feature maps, adaptively graft the essential features between branches, and retain the peculiar features of each branch. It improves the utilization and discrimination of features and reduces the overall amount of calculation.

The remainder of this article is organized as follows. Section II describes the work related to the proposed methods. Section III introduces the proposed methods in detail. Section IV describes the datasets used in this article and the experimental results. Section V concludes this article.

II. Related Work

Before introducing the related work, we make conventions for the notation associated with the MS and PAN images. We take MS for an example and denote it as $M_i^{\exp}(x, y)$, where the superscript is the description of notation. As in this case, exp means example. The subscript i means the ith channel

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of the MS, and (x, y) means the xth row and yth column element.

When representing the entirety, the associated identifiers can be omitted. For example, $M^{\exp}$ represents the entire MS, $M_i^{\exp}$ represents the entire ith channel of MS, and so on. The PAN is denoted as $P_i^{\exp}(x, y)$ with a similar meaning. Unless otherwise stated, the relevant notations appearing next all follow this notation rule.

A. IHS Pansharpening Method

The IHS pansharpening method [41], [42] is one of the most typical CS methods. Initially, the IHS algorithm can only apply to three-channel images. After Tu et al. [11], [43] proposed the GIHS algorithm and its fast algorithm FIHS, respectively, the method can apply to images with any number of channels. Moreover, there is no need to calculate the matrix multiplication and inversion in the GIHS algorithm, which provides great convenience for the fusion.

Under normal circumstances, $M^{\text{raw}} \in \mathbb{R}^{W\times H\times 4}$ and $P^{\text{raw}} \in \mathbb{R}^{4W\times 4H\times 1}$. The GIHS algorithm upsamples the MS image from $M^{\text{raw}} \in \mathbb{R}^{W\times H\times 4}$ to $M^{\text{up}} \in \mathbb{R}^{4W\times 4H\times 4}$ and fuses it using the following equation:

$$M_i^{\text{fuse}} = M_i^{\text{up}} + (P^{\text{raw}} - I_M) \tag{1}$$

where $I_M \in \mathbb{R}^{4W\times 4H\times 1}$ is the intensity component of M^{up}, which is the average of the four channels.

The above methods can obtain good fusion results with high computational efficiency, but there exists a problem of spectral distortion. One of the reasons is that the differences between I_M and P^{raw} are large [44].

To alleviate the above problem, Rahmani et al. [12] proposed the AIHS algorithm. Based on the assumption that the more similar the PAN image with intensity component I_M, the less severe the spectral distortion, and this study introduces a linear factor $\alpha \in \mathbb{R}^4$ to compute the intensity component I_M instead of the mean value

$$I_M = \sum_{i=1}^{4} \alpha_i M_i^{\text{up}} \tag{2}$$

where α can be obtained by optimizing the following convex problem, which minimizes the differences between I_M and P^{raw}

$$\min_{\alpha_i} \left\| \sum_{i=1}^{4} \alpha_i \ M_i^{\text{up}} - P^{\text{raw}} \right\|_2^2 \quad \text{s.t. } \alpha_i > 0. \tag{3}$$

In addition, they also introduce an edge detection operator $W_P \in \mathbb{R}^{4W\times 4H\times 1}$ to extract the edge information of PAN

$$W_P = \exp\left(-\frac{\lambda}{|\nabla P^{\text{raw}}|^4 + \epsilon} \right) \tag{4}$$

where λ is the parameter used to control the gradient size, and it is recommended to set to $\lambda = 10^{-9}$. ϵ is an offset that makes a nonzero denominator, usually set to $\epsilon = 10^{-10}$. ∇P^{raw} is the gradient of the PAN image.

To achieve better fusion, the edge detection operator W_P is used to smooth (1)

$$M_i^{\text{fuse}} = M_i^{\text{up}} + W_P(P^{\text{raw}} - I_M). \tag{5}$$

Leung et al. [13] proposed the IAIHS algorithm to make a further improvement. In addition to the PAN operator, they also proposed an MS edge detection operator $W_{M_i} \in \mathbb{R}^{4W\times 4H\times 1}$ as follows:

$$W_{M_i} = \exp\left(-\frac{\lambda}{\left|\nabla M_i^{\text{up}}\right|^4 + \epsilon} \right) \tag{6}$$

where ∇M_i^{up} is the gradient of the ith channel of M^{up} and λ and ϵ have the same meaning and value as that in (4).

Based on the above two operators, they propose a weighted edge detection operator W_i as follows:

$$W_i = \frac{M_i^{\text{up}}}{1/4\sum_{i=1}^{4} M_i^{\text{up}}} \left(\rho W_{M_i} + (1-\rho) W_P \right) \tag{7}$$

where $\rho \in \mathbb{R}$ is a hyperparameter to regulate the effect of the two edge detection operators.

The fusion result is smoothed in a similar way with (5)

$$M_i^{\text{fuse}} = M_i^{\text{up}} + W_i(P^{\text{raw}} - I_M). \tag{8}$$

With the contributions of researchers, the IHS-series algorithms enjoy wide applications for their fast processing speed and good spatial feature retention. However, the problem of spectral distortion still exists, and hyperparameters in some algorithms also reduce the generalizability. The IHS-series algorithms still have a large room for improvement.

B. Triplet Loss

Schroff et al. [45] proposed a system called FaceNet, which learns the mapping of face images to 128-D Euclidean space by a convolutional neural network, solving the problems of recognition, verification, and clustering of face images based on the similarity between feature vectors. It is worth emphasizing that the study creatively proposes the triplet loss, which is still widely applied today. The triplet loss function is shown as follows:

$$L_t = \max(d(\text{Anc}, \text{Pos}) - d(\text{Anc}, \text{Neg}) + \sigma, 0) \tag{9}$$

where Anc means the anchor, Pos means the positive, and Neg means the negative. σ is the margin and $d(x, y)$ is the Euclidean distance between x and y.

Triplet loss attempts to supervise a feature space where anchor examples are close to positive examples and far away from negative examples. This property makes the neural network with triplet loss distinguish details well, especially in image classification tasks. In other words, when two inputs are similar, triplet loss can supervise a distinguishable representation of them, improving the performance of classification networks.

The deep neural network with triplet loss has excellent nonlinear modeling ability and owns the simplicity of metric learning. It can control the model training process simply and effectively. Although triplet loss has disadvantages such

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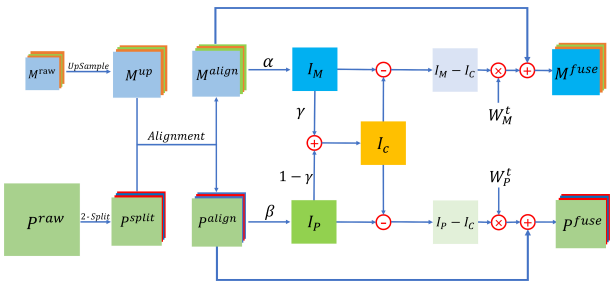

Fig. 1. Flowchart of the ATIHS data fusion strategy. α: linear factor of MS. β: linear factor of PAN. γ: adaptive soft-average mask. I_M: intensity component of MS. I_P: intensity component of PAN. I_S: shared intensity component. W_M^t: weighted operator of MS. W_P^t: weighted operator of PAN.

as increasing the amount of calculation and slowing the convergence in some cases [46], it does not prevent it from being an excellent design.

III. Proposed Method

A. ATIHS Data Fusion Strategy

As previously mentioned, there are representational and structural differences between the MS and PAN images. For the representational differences, the IHS-series algorithms can quickly mitigate them. However, the effect of mitigation mainly depends on the differences between substitution components. How to reduce the differences deserves further study. For the structural differences, the IHS-series algorithms usually upsample the MS image to align the spatial size. However, no further operations are performed on the channel, and subsequent studies also barely consider this. In addition, the IHS-series algorithms aim at pan sharpening, outputting only one fused image. It may not be so suitable for the two-branch network, the mainstream network in this field. Therefore, we propose the ATIHS data fusion strategy, which alleviates the above problems. The whole process of the ATIHS strategy is shown in Fig. 1, which can be divided into five steps as follows.

1) Unification of Image Structure: Generally, the size of the PAN image is four times that of the MS image, and PAN is the single-channel image, while MS is the four-channel image. We denote them as $P^{\text{raw}} \in \mathbb{R}^{4W\times 4H\times 1}$ and $M^{\text{raw}} \in \mathbb{R}^{W\times H\times 4}$. The differences in the structure limit the fusion approach, so we want to adjust the MS and PAN image to the same size and the same number of channels.

For the PAN image, we propose a q-Split operation as follows to make an adjustment:

$$\begin{aligned} &P_i^{\text{split}}(x,y) \\ &= P^{\text{raw}}(qx-q+(i-1)\%q+1, qy-q+\lfloor (i-1)/q \rfloor +1) \\ &\quad i=1,2,\ldots,q^2, \quad x=1,2,\ldots,W'/q, \quad y=1,2,\ldots,H'/q \end{aligned} \tag{10}$$

where q is the split scale, and an example of 2-Split is shown in Fig. 2. W' and H' are the width and height of the image, respectively, which are divisible by q. i, x, and y have been uniformly explained at the beginning of Section II.

In fact, each channel extracted by the q-Split operation can be regarded as q^2 different downsampling subimages. Each subimage can well represent the scene of the original image. In more depth, q-Split maps the spatial feature into the spectral-like feature, enabling the PAN images to expand to q^2 channels with a strong interchannel correlation. For the PAN image, we perform a 2-Split on it, after which PAN changes from $\mathbb{R}^{4W\times 4H\times 1}$ to $\mathbb{R}^{2W\times 2H\times 4}$. Correspondingly, for the MS image, we perform a twofold upsampling, turning it from $\mathbb{R}^{W\times H\times 4}$ to $\mathbb{R}^{2W\times 2H\times 4}$. At this time, they have the same shape, and we denote them as P^{split} and M^{up}.

Though M^{up} and P^{split} have the same shape now, the order of their channels is not necessarily aligned. Therefore, we would like to adjust the channel order by channel similarity between the MS and PAN images. In order to achieve the above purpose, we calculate the Gram matrix, which represents the similarity between channels

$$G(u,v) = \sum_{x=1}^{2W}\sum_{y=1}^{2H} M_u^{\text{up}}(x,y) P_v^{\text{split}}(x,y) \tag{11}$$

where $G \in \mathbb{R}^{4\times 4}$ and $G(u,v)$ are the uth row and the vth column element of the Gram matrix, respectively. A larger value of $G(u,v)$ means larger similarity between the uth channel of M^{up} and the vth channel of P^{split}.

Then, we need to find an alignment solution that maximizes the channel similarity between the MS and PAN images. All feasible solutions can be obtained by solving the 4-Queens Problem (diagonal allowed) on the Gram matrix. The solution with the largest summation value is selected as the final alignment solution, in which case M^{up} has the largest similarity with P^{split}. After alignment, their structural differences are further reduced, and we denote them as $M^{\text{align}} \in \mathbb{R}^{2W\times 2H\times 4}$ and $P^{\text{align}} \in \mathbb{R}^{2W\times 2H\times 4}$.

2) Intensity Components of the MS and PAN Image: In the IHS-series algorithms, the MS intensity component $I_M \in \mathbb{R}^{4W\times 4H\times 1}$ is defined in (2), and it is considered that $I_P = P \in \mathbb{R}^{4W\times 4H\times 1}$ since PAN is the single-channel image. In our work, the definition of I_M is the same, but that of I_P changes. Because the 2-Split operation makes the PAN image own four channels, the definition of our I_P is similar to I_M now. Moreover, the shape of them also changes, where $I_M \in \mathbb{R}^{2W\times 2H\times 1}$ and $I_P \in \mathbb{R}^{2W\times 2H\times 1}$

$$I_M = \sum_{i=1}^{4} \alpha_i M_i^{\text{align}} \tag{12}$$

$$I_P = \sum_{i=1}^{4} \beta_i P_i^{\text{align}} \tag{13}$$

where $\alpha \in \mathbb{R}^4$ and $\beta \in \mathbb{R}^4$ are linear factors.

Correspondingly, we rewrite the convex optimization problem of (3). By optimizing it, we can obtain α and β that minimize the differences between I_M and I_P

$$\min_{\alpha_i,\beta_i} \left\| \sum_{i=1}^{4} \alpha_i M_i^{\text{align}} - \sum_{i=1}^{4} \beta_i P_i^{\text{align}} \right\|_2^2$$

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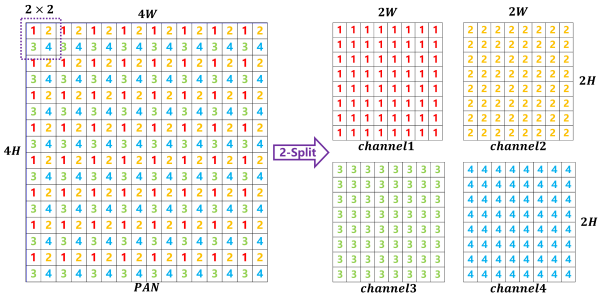

Fig. 2. Illustration on the 2-Split operation. The PAN image is divided into 2×2 patches, and then, pixels at the same position in each patch are extracted as a channel of the new image.

$$\text{s.t. } \alpha_i, \beta_i > 0, \quad \sum_{i=1}^{4} \alpha_i = 1. \tag{14}$$

3) Shared Intensity Component: We introduce a new concept, defining the component used for substitution as the shared intensity component I_S. On this premise, we have $I_S = I_P = P$ in the IHS-series algorithms. However, we believe that replacing I_M with $I_S = \text{PAN}$ is a bit simple and rigid, and there is still a nonignorable gap between them and can be further improved. Therefore, we novelly propose I_S as follows:

$$I_S = \gamma\, I_M + (1-\gamma) I_P. \tag{15}$$

In our opinion, I_S is supposed to be a weighted sum of I_M and I_P so as to fully combine their characteristics and further reduce the gap between replacement components. Moreover, we are not limited to simple linear weighting, further proposing an adaptive soft-average mask $\gamma \in \mathbb{R}^{2W\times 2H\times 1}$ as follows:

$$\gamma = \frac{1}{1+e^{-[\text{sign}(I_M - I_P)](I_A - I_G)}} \tag{16}$$

where $\text{sign}(\cdot)$ is the signum function and I_A and I_G are the average intensity component and global intensity component, respectively, defined as follows:

$$I_A = \frac{I_M + I_P}{2} \tag{17}$$

$$I_G = \frac{\sum_{x=1}^{2W}\sum_{y=1}^{2H} \frac{I_M(x,y)+I_P(x,y)}{2}}{2W \times 2H}. \tag{18}$$

We assume that $I_G \in \mathbb{R}$ can represent the overall middle plane of I_M and I_P. This plane can divide elements in $I_A \in \mathbb{R}^{2W\times 2H\times 1}$ into two parts: elements biased toward I_M and elements biased toward I_P. We want to further preserve the bias characteristics of each element of I_A, so $[\text{sign}(I_M - I_P)](I_A - I_G)$ is input into the sigmoid function, where $\text{sign}(I_M - I_P)$ is used to adjust the correct bias direction. Weighted by γ, I_S becomes a smoother average between I_A and I_G and preserves more respective characteristics of I_M and I_P.

4) Edge Detection Operators: To further smooth the fusion, we use the edge detection operators to extract the edge information of the MS and PAN image. The basic edge detection operator W_{M_i} has been defined at (6), while W_{P_i} is rewritten due to the q-Split operation

$$W_{M_i} = \exp\left(-\frac{\lambda}{\left|\nabla M_i^{\text{align}}\right|^4 + \epsilon}\right) \tag{19}$$

$$W_{P_i} = \exp\left(-\frac{\lambda}{\left|\nabla P_i^{\text{align}}\right|^4 + \epsilon}\right). \tag{20}$$

In order to make the MS and PAN images able to learn each other's edge information and balance the influence of the operators, we propose tradeoff edge detection operators $W_{M_i}^t$ and $W_{P_i}^t$

$$W_{M_i}^t = \frac{\alpha_i}{1/4\sum_{i=1}^{4}\alpha_i}\left(\gamma\, W_{M_i} + (1-\gamma) W_{P_i}\right) \tag{21}$$

$$W_{P_i}^t = \frac{\beta_i}{1/4\sum_{i=1}^{4}\beta_i}\left(\gamma\, W_{M_i} + (1-\gamma) W_{P_i}\right). \tag{22}$$

The coefficient $(\alpha_i)/((1/4)\sum_{i=1}^{4}\alpha_i)$ keeps the ratio among the MS image bands unchanged, and the coefficient $(\beta_i)/((1/4)\sum_{i=1}^{4}\beta_i)$ can be applied to the PAN image symmetrically. Moreover, the above two coefficients are hyperparameter-free, improving the adaptability of the algorithm. Naturally, we assume that the tradeoff of intensity components is similar to the tradeoff of edge detection operators, so they can share the same γ. This consistent treatment of γ is reasonable, making the detection of spatial details smoother.

5) Fusion of the MS and PAN Image: After computing the shared intensity components and the edge detection operators, we can obtain the fused image by the following equation:

$$M_i^{\text{fuse}} = M_i^{\text{align}} + W_{M_i}^t (I_S - I_M) \tag{23}$$

$$P_i^{\text{fuse}} = P_i^{\text{align}} + W_{P_i}^t (I_S - I_P). \tag{24}$$

Overall, the ATIHS strategy effectively reduces structural and representational differences, paving the way for the neural network, especially the two-branch network. On the one hand, the ATIHS strategy unifies an image structure, reducing structural differences effectively. On the other hand, the synchronous optimization of α and β coefficients further reduces the gap between replacement components than single coefficient α. Moreover, the introduction of γ further reduces the representational differences of M^{fuse} and P^{fuse}. In addition, the ATIHS is a symmetrical fusion method, obtaining two fusion results that can be input to a two-branch network in parallel.

B. FG-Block

The MS and PAN images inherently share the essence that represents the same region, while they own unique peculiarities in the respective spectral and spatial domains. We believe that the similar features extracted from different modal data can reflect the essential features of the scene, while the peculiar features reflect the aspects that each modality is specialized. How to effectively utilize essential and peculiar features is the key to improving the classification performance. However, most studies extract features from MS and PAN entirely,

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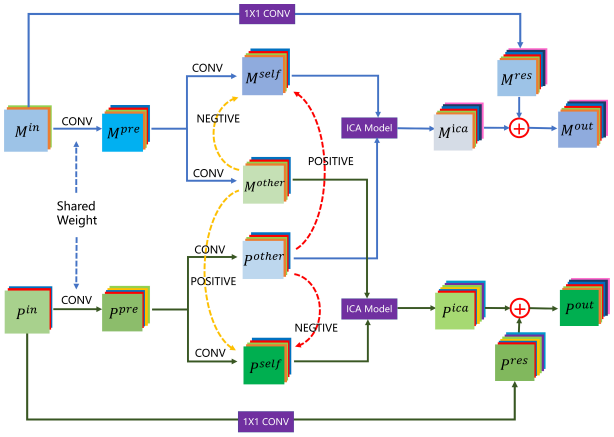

Fig. 3. Flowchart of the FG-Block. Taking the MS branch as an example, the input M^{in} is convoluted to M^{pre}, and whether to share the weights of this part is adjustable. Then, M^{pre} is convoluted to M^{self} and M^{other}, and the detailed triplet relationship is shown in the picture. The ICA module takes M^{self}, P^{other}, and some intermediate results of triplet loss as the input, outputting the M^{ica}. Finally, add M^{ica} and M^{res} to obtain the result M^{out}.

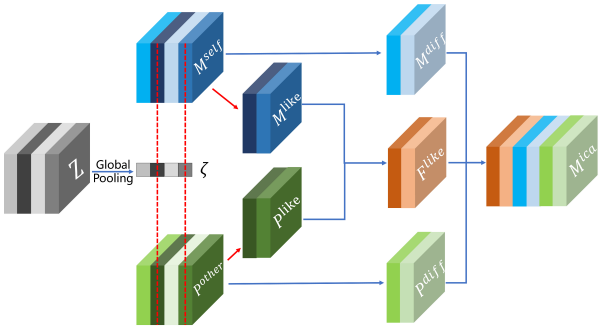

Fig. 4. Illustration on the ICA module. Z: integrated metric tensor. ζ: integrated metric vector. F^{like}: fusion result of M^{like} and P^{like}. M^{ica}: final fusion result.

ignoring the linkage and differences between essence and peculiarity. Therefore, we propose FG-Block, which tries to separate and haul essential and peculiar features within the branch and graft them across the branch. The flowchart of FG-Block is shown in Fig. 3. Since the structure of the FG-Block is symmetrical, for simplicity, we take the MS branch as an example to illustrate the details of the FG-Block.

1) Feature Preprocess: This section is the initial convolutional part of the block, which is used to extract preliminary features and pave the way for further feature extraction. We denote the input of this part as M^{in} and the output as M^{pre}.

2) Feature Separation and Traction: In this section, the preliminary feature M^{pre} is convolved by different convolution blocks. One gets M^{self}, and another gets M^{other}. It achieves the initial separation of features, and we want to make more purposeful traction on the features to achieve better separation. To fulfill our expectations, we introduce a triplet loss [45] to supervise the learning process of M^{self}, which can be calculated as follows:

$$L_t = \max\left(d\left(M^{\text{other}}, P^{\text{self}}\right) - d\left(M^{\text{other}}, M^{\text{self}}\right) + \sigma, 0\right). \tag{25}$$

Supervised by triplet loss, M^{other} would be away with M^{self} to reduce peculiar information, and it would be closer to P^{self} to achieve learning of essential features. It means the separation of the homogeneity and heterogeneity, which improves the discrimination between the two branched feature blocks. For M^{self}, it should not be additionally supervised by a triplet loss to maintain its stability as an object to be learned.

3) Feature Mutual Graft by ICA Module: Now, we have M^{self}, M^{other}, P^{self}, and P^{other} four feature blocks. To graft the essence features learned from the opposite branch, we want M^{self} to be fused with P^{other} and P^{self} to be fused with M^{other}. The general approach is elementwise adding and concatenating, but these two operations are simple and rigid. The former easily loses information, while the latter is prone to cause information redundancy. Therefore, we propose an ICA module, and the basic process is shown in Fig. 4. We still take the MS branch as an example, and the details are explained as follows.

First, we propose Z called integrated metric tensor, which is the intermediate results of (25)

$$Z = \left|M^{\text{other}} - M^{\text{self}}\right| - \left|M^{\text{other}} - P^{\text{self}}\right|. \tag{26}$$

Second, global pool the integrated metric tensor Z to obtain the integrated metric vector ζ. According to the definition, the larger ζ_i is, the higher the similarity between M_i^{other} and P_i^{self}. Then, the top-K channels with the largest value in ζ, i.e., the K channels with the most similarity between M_i^{other} and P_i^{self} are selected adaptively. Separate top-K channels from M^{other} and P^{self} into M^{like} and P^{like}, and recombine the rest as M^{diff} and P^{diff}.

Third, M^{like} and P^{like} are fused to obtain F^{like}. The fusion method can be a simple average or the weights can be adaptively learned by a neural network module.

Finally, concatenate F^{like}, M^{diff}, and P^{diff} to obtain the final fusion result M^{ica}.

This method is a tradeoff between elementwise adding and concatenating. It degenerates to elementwise adding when K is all channels and degenerates to concatenating when K is 0. The ICA module mitigates the modal inconsistency and information loss in elementwise adding and alleviates the sharp computation increase and information redundancy caused by concatenating.

4) Feature Fusion: To further improve the network performance, we introduce residual path in each branch, and the initial inputs M^{in} and P^{in} are processed by 1×1 convolution layers to obtain M^{res} and P^{res}, respectively, which are added with M^{ica} and P^{ica} to obtain the final fusion results M^{out} and P^{out}.

The FG-Block fully exploits the homogeneity and heterogeneity of remote sensing images, achieving the separation and fusion of essential and peculiar features through two symmetrical branches. The ICA module mutually grafts essential and peculiar features, and it realizes adaptive channel selection, suppressing unnecessary dimensional expansion effectively. Moreover, the reuse of intermediate results further reduces the computational burden.

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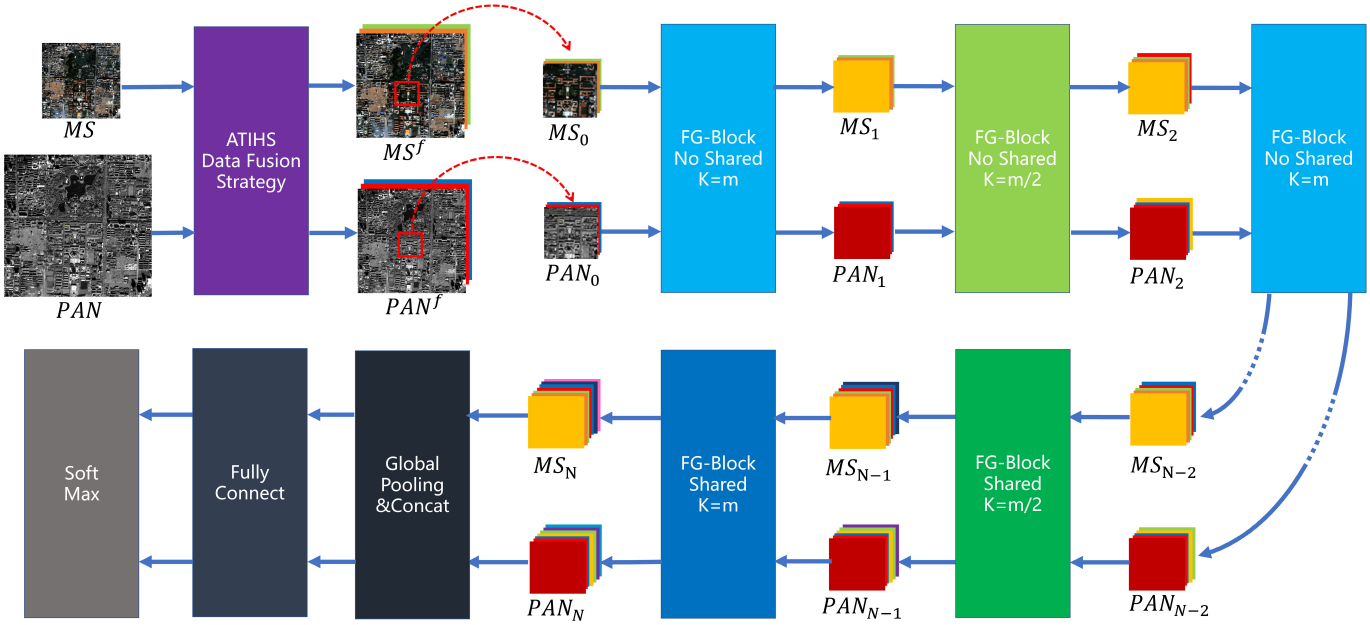

Fig. 5. Whole proposed architecture of TSMF-Net. The whole process consists of three parts: 1) data fusion through the ATIHS strategy; 2) feature fusion net consists of N repetitive FG-Blocks with different parameters; and 3) pooling, concatenation, full connection, and classification.

C. Construction of the TSMF-Net

In this section, we describe our proposed TSMF-Net in detail. The architecture of TSMF-Net is shown in Fig. 5 and can be divided into the following three parts.

1) Data Fusion: Before feeding into the neural network to train, we use the ATIHS strategy to fuse the original MS and PAN image to reduce the differences between them, paving the way for feature extraction.

2) Feature Fusion: We cut the fusion results MS^f and PAN^f into patches as the input of the net. This section consists of N repetitive FG-Blocks, while there are some differences in their parameters. In our proposed network, the weights in the feature preprocess part of the first $\lfloor (N/2) \rfloor$ FG-Blocks are not shared, while the weights of the rest FG-Blocks are shared. This is because the features extracted by the shallow blocks represent specific details, and sharing weights would lead to confusion and weaken the discriminative ability of the network. However, the features extracted by the deeper blocks represent similar semantic information. At this time, sharing weights would not lead to confusion, and it can reduce excessive model parameters and amount of calculation. For simplicity, there are only two choices of K in the ICA module of FG-Block, $(m/2)$ and m, where m is the channel numbers. When $K = (m/2)$, the ICA module takes the most similar half of all channels for elementwise adding, and the rest are concatenated, making the fusion of features balanced. When $K = m$, the ICA module degenerates to elementwise adding. The block with $K = m/2$ can purposefully fuse the information, but its top-K ordering may bring some uncertainty, so we need the block with $K = m$ as a transition to improve the stability of the network. Therefore, we set the above two K values to FG-Block alternately, and K of the N FG-Blocks is taken as $m, (m/2), m, (m/2), m, \ldots$

3) Feature Classification: Finally, we obtain the feature block MS_N and PAN_N. After passing through a global pooling layer and several fully connected layers, the class probability is finally estimated.

The overall loss of the network consists of two parts: the first part is the cross-entropy loss and the second part is the weighted sum of the triplet loss of each FG-Block. Compared with multitask learning, our model is more oriented toward optimizing for a single task, where the relationship between triplet loss and cross-entropy loss is not completely antagonistic but relatively synergistic

$$L = L_c + \mu \sum_{i=1}^{N} L_{t_i} \tag{27}$$

where L_c refers to the cross-entropy loss, L_{t_i} refers to the triplet loss of the ith FG-Block block, and μ is the weight.

IV. Experimental Results

A. Datasets Description

In this section, we use four datasets to validate the reasonableness and effectiveness of our proposed methods. All of the data are obtained by the GaoFen-2 satellite, whose spatial resolution is 3.2 m and spatial resolution is 0.8 m. The details of the datasets are given as follows.

1) The Hohhot area is shown in Fig. 6(a). The MS image contains 2001 × 2101 × 4 pixels, and the PAN image has 8004 × 8404 pixels. It can be divided into 11 categories: water, road, bare land, vegetation, farmland, building in three different areas, and buildings with three different roof colors.
2) The Nanjing area is shown in Fig. 6(b). The MS image consists of 2000 × 2500 × 4 pixels. The PAN image

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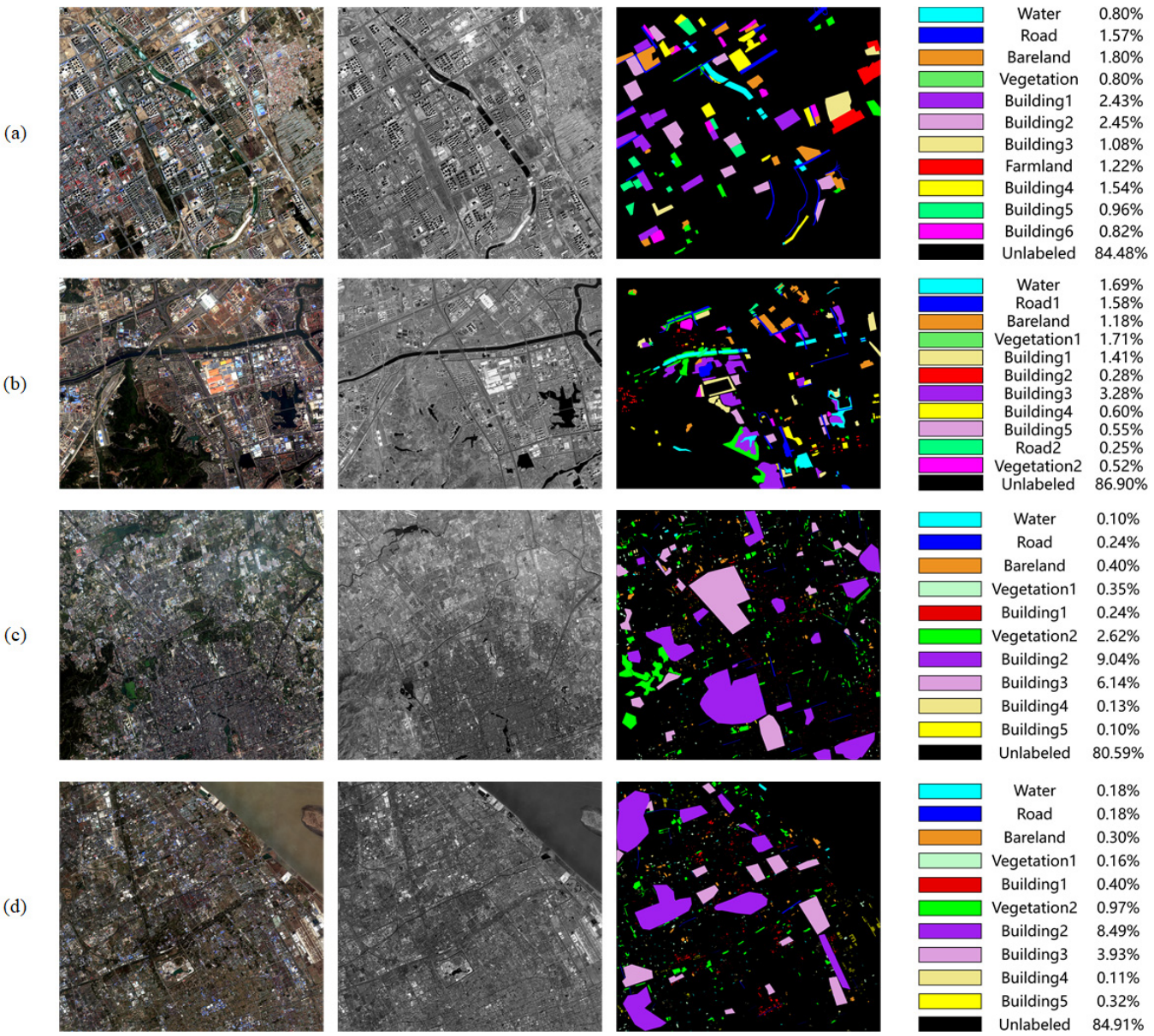

Fig. 6. Four rows in order from top to bottom are (a) Hohhot area datasets, (b) Nanjing area datasets, (c) Beijing area datasets, and (d) Shanghai area datasets. The four columns from left to right are the MS images, PAN images, ground-truth images, and corresponding labels. In addition, some images are resized to obtain better visibility.

owns 8000 × 10 000 pixels. The dataset is segmented into 11 types: water, bare land, two types of road, two kinds of vegetation, and five kinds of buildings.

3) The Beijing area is shown in Fig. 6(c). The MS image is made up of 6908 × 7300 × 4 pixels, while the PAN image is formed by 27 620 × 29 200 pixels, which can be grouped into ten categories: water, road, bare land, two species of vegetation of different heights, and five kinds of buildings.
4) The Shanghai area is shown in Fig. 6(d). The MS image contains 6908 × 7300 × 4 pixels, while the PAN image owns 27 620 × 29 200 pixels. It is divided into ten categories: water, road, bare land, two species of vegetation varying in height, and five types of buildings.

B. Experimental Environment and Settings

The experiments in this article are conducted on a workstation with RTX2080Ti 12-GB GPU and 128-GB RAM under Ubuntu 16.04 LTS. All experiments are done under the PyTorch framework, except the linear factors α and β in (14), which are computed by MATLAB due to the huge size of the matrix.

In order to measure the rationality and efficiency of our proposed method, we use indicators, such as overall accuracy (OA), average accuracy (AA), and kappa statistics (KAPPA), to quantitatively analyze our work. The detailed network parameters of our proposed TSMF-Net are shown in Table I. The patch size of both MS and PAN is 32 × 32 × 4.

For the network training, we randomly select 2% labeled data from every class as training samples. In order to make

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TABLE I
HYPERPARAMETERS OF THE TSMF-NET

Stage		Type	Input Size	Filter Size	stride & padding	Output Size
Data Fusion		ATIHS Strategy	W×H, 4(MS) 4W×4H, 1(PAN)	None		2W×2H, 4
		Patch cutting	2W×2H, 4	None		32×32, 4
The ith FG-Block in the Feature Fusion (i=1,2...N)	odd	$Conv_M^{pre}$, $Conv_P^{pre}$	32×32, $4 \times 6^{(i-1)/2}$	3×3, $8 \times 6^{(i-1)/2}$	1, 1	32×32, $8 \times 6^{(i-1)/2}$
		$Conv_M^{self}$, $Conv_P^{other}$	32×32, $8 \times 6^{(i-1)/2}$	3×3, $8 \times 6^{(i-1)/2}$	1, 1	32×32, $8 \times 6^{(i-1)/2}$
		$Conv_P^{self}$, $Conv_M^{other}$	32×32, $8 \times 6^{(i-1)/2}$	3×3, $8 \times 6^{(i-1)/2}$	1, 1	32×32, $8 \times 6^{(i-1)/2}$
		ICA_M, ICA_P	32×32, $8 \times 6^{(i-1)/2}$	None		32×32, $8 \times 6^{(i-1)/2}$
		$Conv_M^{res}$, $Conv_P^{res}$	32×32, $4 \times 6^{(i-1)/2}$	1×1, $8 \times 6^{(i-1)/2}$	1, 0	32×32, $8 \times 6^{(i-1)/2}$
	even	$Conv_M^{pre}$, $Conv_P^{pre}$	32×32, $8 \times 6^{(i-2)/2}$	3×3, $16 \times 6^{(i-2)/2}$	1, 1	32×32, $16 \times 6^{(i-2)/2}$
		$Conv_M^{self}$, $Conv_P^{other}$	32×32, $16 \times 6^{(i-2)/2}$	3×3, $16 \times 6^{(i-2)/2}$	1, 1	32×32, $16 \times 6^{(i-2)/2}$
		$Conv_P^{self}$, $Conv_M^{other}$	32×32, $16 \times 6^{(i-2)/2}$	3×3, $16 \times 6^{(i-2)/2}$	1, 1	32×32, $16 \times 6^{(i-2)/2}$
		ICA_M, ICA_P	32×32, $16 \times 6^{(i-2)/2}$	None		32×32, $4 \times 6^{(i/2)}$
		$Conv_M^{res}$, $Conv_P^{res}$	32×32, $8 \times 6^{(i-2)/2}$	1×1, $4 \times 6^{(i/2)}$	1, 0	32×32, $4 \times 6^{(i/2)}$
Classification	odd	Pooling&Concat	32×32, $8 \times 6^{(N-1)/2}$	None		$2 \times 8 \times 6^{(N-1)/2}$
		Fully Connect	$2 \times 8 \times 6^{(N-1)/2}$	None		class
	even	Pooling&Concat	32×32, $4 \times 6^{(N/2)}$	None		$2 \times 4 \times 6^{(N/2)}$
		Fully Connect	$2 \times 4 \times 6^{(N/2)}$	None		class

TABLE II
EXPERIMENTAL RESULTS OF HYPERPARAMETER BLOCK NUMBERS N (%)

Dataset	Hohhot Area			Nanjing Area			Beijing Area			Shanghai Area		
FG-Block Numbers	3	4	5	3	4	5	3	4	5	3	4	5
OA	97.39	99.17	98.33	96.91	99.02	98.55	92.24	94.03	93.70	91.24	93.78	93.25
AA	97.07	99.16	98.22	97.26	99.18	98.53	86.44	92.89	88.07	87.44	93.74	91.53
KAPPA	97.07	99.07	98.13	96.42	98.87	98.32	88.43	91.14	90.57	85.60	90.04	88.94

the training samples balanced, for the categories with less than 1500 samples, we supplement the samples to 1500. When selecting the test set, we set the intersection-over-union (IOU) ratio as the guideline. Samples with the IOU ratio greater than 1/3 with other training samples will be excluded. The initial training rate is 6×10^{-4}, the batch size is 64, and the weight decay is 5×10^{-4}. To reduce the influence of random factors and improve experimental reliability, we repeat experiments five times and take the average value as the final result.

C. Hyperparameter Analysis and Selection

In this section, we perform a detailed quantitative analysis of the hyperparameters using the control variable method. We leave the parameters identical except for the selected hyperparameters.

1) Comparison and Analysis of FG-Block Numbers N: As far as we are concerned, when the number of blocks is too small, the network has difficulty in extracting deep features and has poor expressiveness. When the number of blocks is too large, network parameters increase sharply, and overfitting easily occurs, reducing the classification performance. Table II shows the quantitative comparison of the FG-Block numbers on the different datasets, and we can easily find that the network performance is best when the number of blocks is 4. Due to the discrete nature of the number of blocks, each addition or subtraction of a block can significantly impact the network performance. Therefore, we can conclude that the best block number is 4, and the following experiments will be conducted on the four-block TSMF-Net.

2) Exploration of the Values of σ and μ: σ is the margin of triplet loss, and μ is the weight of the triplet loss in (27). Table III shows the grid search results for σ and μ on the Hohhot dataset. The network has the best performance when $\sigma = 0.3$ and $\mu = 0.25$. For μ, moderate μ can control the effect of triplet loss in the network, making an appropriate convergence. For σ, we take the MS branch as an example. When σ is too small, M^{other} is too similar to P^{self}, causing loss of information and degradation of the network performance. Conversely, when σ is too large, its learning ability becomes poor, and it cannot fully learn from the other branch, causing the network's poor performance.

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TABLE III
EXPERIMENTAL RESULTS OF HYPERPARAMETER σ AND μ ON THE HOHHOT DATASET (%)

σ \ μ	0.05			0.15			0.25			0.35			0.45		
0.1	98.72	98.63	98.57	98.98	98.91	98.85	98.78	98.67	98.63	98.85	98.76	98.72	98.88	98.77	98.74
0.2	98.90	98.85	98.78	98.85	98.77	98.71	98.97	98.90	98.84	98.93	98.90	98.80	98.78	98.77	98.63
0.3	98.85	98.78	98.71	99.03	98.97	98.92	99.17	99.16	99.07	98.90	98.85	98.77	98.93	98.85	98.80
0.4	98.97	98.93	98.85	98.99	98.89	98.87	99.04	98.98	98.93	98.75	98.68	98.61	98.80	98.62	98.66
0.5	98.71	98.57	98.56	98.70	98.64	98.54	98.74	98.65	98.59	98.82	98.72	98.68	98.85	98.72	98.72

TABLE IV
EXPERIMENTAL RESULTS OF HYPERPARAMETER PATCH SIZE M (%)

Dataset	Hohhot Area			Nanjing Area			Beijing Area			Shanghai Area		
Patch Size	24	32	40	24	32	40	24	32	40	24	32	40
OA	97.88	99.17	99.17	97.62	99.02	98.47	92.76	94.03	94.31	89.32	93.78	94.65
AA	97.69	99.16	99.11	97.68	99.18	98.42	89.13	92.89	90.86	86.03	93.74	90.25
KAPPA	97.62	99.07	99.07	97.23	98.87	98.23	89.26	91.14	91.53	82.80	90.04	91.20
Time	346.58	427.89	535.79	374.21	391.33	569.88	3006.57	3542.14	3949.75	2956.31	3298.26	3788.96

3) Comparison and Exploration of Patch Size M: As shown in the Table IV, when the patch size is small, although the running time decreases a lot, the metrics of classification decrease across the board. When the patch size is large, the running time increases accordingly. The OA is also improved on some datasets. However, the stability of the classification decreases. In fact, it is closely related to image resolution and object size. Small-size patches cannot completely contain the target object, and loss of information will reduce classification accuracy and stability. Large-size patches can effectively include the targets, but it also introduces redundant interference information at the edge of the patch. It can improve the classification accuracy in some categories but will reduce the overall stability of the classification. Considering the time cost and classification performance, choosing the patch size of 32×32 is reasonable and effective.

D. Analysis of the ATIHS Strategy

In this section, we take the Hohhot area dataset as an example, analyze our proposed ATIHS algorithm, and compare it with other IHS-series algorithms. For experimental fairness, all the algorithms use the edge detection operators of the same parameters and the TSMF-Net trained with the same number of iterations. To ensure the comprehensiveness of the validation, we compare the performance of each algorithm in two stages: during fusion and after fusion.

1) During Fusion: One of the most important steps in the fusion process is the calculation of intermediate components. It usually includes I_M, I_P, and I_S three components, representing the intensity component of MS, the intensity component of PAN, and the shared intensity component. These intermediate components will be used for CS, and the smaller differences between them, the smaller the negative impact of the substitution will bring. Therefore, we conduct an experiment on the differences in the components and introduce mean square error (mse) as the evaluation metric. To facilitate comparison, we let $I_S = I_A$ in IHS-series algorithms, and I_A is also compared in the table.

TABLE V
MSE BETWEEN THE INTERMEDIATE COMPONENTS ON THE HOHHOT AREA DATASET

MSE($\times 10^{-5}$)	I_M	I_P	I_S	I_A
I_M		50.012	13.758	13.758
I_P	32.744		13.758	13.758
I_S	8.050	8.415		0.000
I_A	8.186	8.186	0.047	

The results are shown in Table V. Since the comparison results of both the ATIHS and IHS-series algorithms are symmetric matrices with a diagonal of 0, we integrate them into one matrix for the convenience of comparison. The lower half of Table V is the result of the ATIHS strategy, and the upper half is the result of IHS-series algorithms.

We can see that the mse between intermediate components of the ATIHS strategy is significantly smaller than that of IHS-series algorithms, where the former is nearly 40% smaller than the latter. We also found that our I_S has an unequal bias with I_M and I_P while having tiny differences with I_A, indicating that I_S retains more information specific to I_M and I_P without losing balance. Thus, we can say that the ATIHS strategy can effectively reduce the differences between components while retaining more specific information in balance.

2) After Fusion: After the MS and PAN images complete fusion, we visualize them for qualitative comparison, where Fig. 7 shows the results. Here, we take the AIHS, IAIHS, and our ATIHS for comparison.

For the MS images, the fused MS images obtained by the AIHS and IAIHS algorithms show a distinct color distortion. The image color is richer in blue and gray, indicating spectral distortion to some extent. In contrast, the image obtained by the ATIHS algorithm has a very similar color tone to the original MS image, demonstrating that our algorithm effectively alleviates the spectral distortion. In addition, the edges of streets and houses in some areas are moderately

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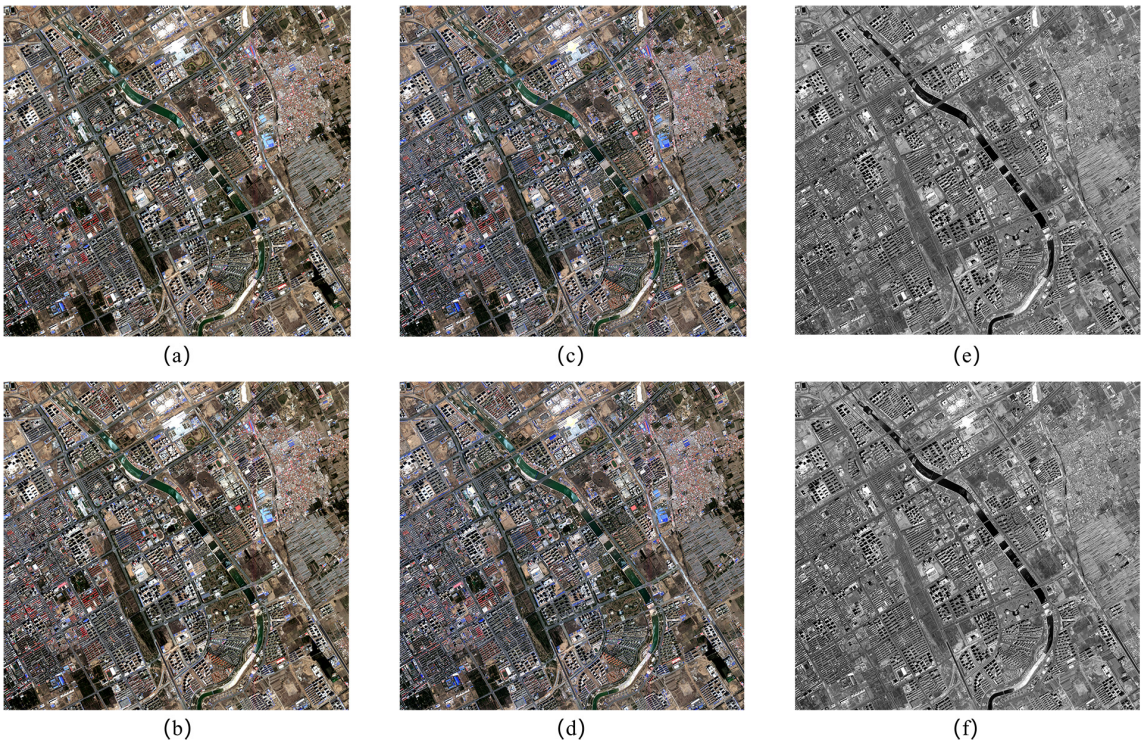

Fig. 7. Fusion results. (a) Original MS image. (b) ATIHS processed MS image. (c) AIHS processed MS image. (d) IAIHS processed MS image. (e) Original PAN image. (f) ATIHS processed PAN image. Some images are resized to obtain better visibility.

sharp, and the color transition is softer, better reproducing the real scene.

For the PAN images, the AIHS and ATIHS do not have the corresponding PAN image fusion results because they are just pan-sharpening algorithms, so here, we only compare the original PAN image with the ATIHS fused PAN image. We find that the grayscale transition of the ATIHS fused PAN image is smoother than the original one. There are smoother transition bands next to some edges, which indicates that the PAN image fuses the spectral information from the MS image, further proving the effectiveness of our algorithm.

Finally, we feed the fused MS and PAN images into the network to quantify the performance of the three algorithms, and the results are shown in Table VI.

Our algorithm obtains the highest OA, AA, and KAPPA, and it has higher accuracy than the other two algorithms in most categories. It proves the reasonableness and progressiveness of our algorithm to some extent. It is worth noting that AIHS achieves the highest accuracy on c2 (Road). This is because AIHS only introduces the edge detection operator of PAN without excessive smoothing operations, which allows it to retain more spatial information and achieve better performance in the categories with complex topography.

TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT DATA FUSION STRATEGIES ON THE HOHHOT AREA DATASET (%)

Strategy	AIHS	ATIHS	IAIHS
c_1	99.07	99.68	99.21
c_2	99.14	97.72	96.02
c_3	94.65	99.32	99.12
c_4	97.80	99.34	98.67
c_5	96.45	99.23	99.27
c_6	98.90	99.57	98.19
c_7	96.65	99.40	99.32
c_8	99.72	99.50	98.21
c_9	96.89	99.34	99.37
c_{10}	99.13	99.18	98.31
c_{11}	97.08	98.49	93.50
OA	97.63	99.17	98.28
AA	97.77	99.16	98.11
KAPPA	97.34	99.07	98.08

E. Analysis of ICA Module

In this section, we make a specific comparison between elementwise adding, concatenating, and our proposed ICA module. We still adopt the control variable method to conduct our experiments.

The experimental results are shown in Table VII. Our proposed ICA module achieves the best results for OA, AA, and KAPPA. Elementwise adding has a small number of parameters, but it tends to lose information. Concatenating can completely retain all the information, but it is easy to cause a sharp increase in size, where the redundant information tends to degrade the network performance. Our proposed ICA module purposefully conducts elementwise adding to the most similar part and conducts concatenating to the other part. It can control the size of feature blocks within a reasonable range, reducing information redundancy while retaining important information.

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TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT FEATURE FUSION ALGORITHMS ON THE HOHHOT AREA DATASET (%)

Algorithm	Element-wise Adding	ICA	Concatenating
c_1	99.00	99.68	99.19
c_2	96.29	97.72	97.37
c_3	98.59	99.32	97.50
c_4	95.62	99.34	96.83
c_5	97.55	99.23	98.47
c_6	98.82	99.57	99.59
c_7	97.97	99.40	99.06
c_8	98.89	99.50	98.34
c_9	96.73	99.34	98.25
c_{10}	97.13	99.18	97.44
c_{11}	94.09	98.49	96.58
OA	97.57	99.17	98.22
AA	97.33	99.16	98.05
KAPPA	97.28	99.07	98.01

F. Experimental Result With Beijing Area Dataset

In this section, to test our proposed algorithm's effectiveness, we take the Beijing area dataset as an example, conducting experiments to compare our algorithm with other algorithms. The algorithms include adaptive center-offset sampling strategy (ACO-SS) + DBFA-Net [47], AHF-Net [39], GAFnet [48], CRHFF [49], and MFT [50]. In order to ensure the fairness of the experiments, all parameters are set to the best parameters in their original paper.

The qualitative and quantitative experimental results on the Beijing area dataset are shown in Fig. 8 and Table VIII, and the experimental results of different algorithms are analyzed as follows.

ACO SS + DBFA-Net's ACO-SS, SA-module, and CA-module effectively enhance network adaptability in the spatial and spectral domains. However, for regions like c1 (Water), where there are no other targets around, the strategy shifts all pixels, reducing the classification accuracy.

AHF-Net improves the classification accuracy in most categories compared to ACO-SS + DBFA-Net. It proposes a new fusion strategy AWIHS, which further improves the classification accuracy in spectrum information-rich regions like c1 (Water). Besides, it applies the thought of the feature pyramid network to construct the network, which improves the classification accuracy of the area with varied geography.

The overall metrics of GAFnet are similar to that of AHF-Net, but GAFnet has a better performance in c1 (Water) and c8 (Buding3). In particular, c1 (Water) is further improved than AHF-Net and achieves the highest accuracy among all algorithms. It is because GAFnet fully integrates multilevel features, considers both low-level features and high-level features, and maintains local detailed and global abstract information. However, the parameters of GAFnet are an order of magnitude larger than other algorithms. The excessive prediction time may lead to difficulties in practical applications.

Compared with the previous methods, CRHFF improves the classification accuracy in most categories and especially achieved excellent results in the building categories. CRHFF can extract and fuse cross-resolution potential information, better model the imaged objects from multimodal representations, and significantly improve the classification accuracy of complex categories. The classification accuracy of building categories c5, c7, c8, c9, and c10 is significantly improved, and in particular, the more complex and diverse c7 and c9 categories achieve the highest accuracy.

The MFT makes full use of multimodal data and achieves the satisfactory experimental results. Compared with CRHFF, the overall OA of MFT is much improved. Although it does not perform well on c2 (Road), which has many surrounding categories, it improves over CRHFF on several other categories, especially achieving the highest accuracy on c7 (Building2), with an improvement of more than two percentage points.

Our proposed TSMF-Net has the best overall performance compared to all previously mentioned methods. For the nonbuilding categories except c1 (Water), our algorithm achieves the highest accuracy because it considers both essential and peculiar features and separates, strengthens, and fuses them effectively. Therefore, our network has good expressiveness and discrimination ability to effectively classify multiple categories. For the building categories, our algorithm is weaker than CRHFF on c7 and c9, but the differences are not significant. However, our algorithm is stronger than CRHFF on c8 (Buding3) and c10 (Buding5), and especially on c10, the classification accuracy improves by nearly 8%. It is because the spatial information of buildings is similar, but the spectral characteristics of buildings are different as they are composed of different materials. Our proposed TSMF-Net fuses spectral and spatial information mutually. Although the domain information is unbalanced, the complementary fusion results can still be easily distinguished. Overall, our model is more stable than MFT, has fewer parameters, and performs well on a wider range of data than CRHFF.

ACO-SS + DBFA-Net has the least number of parameters. However, the ACO-SS needs to compute the DoG features and Voronoi diagram when generating patches, which requires considerable computation and leads to high time costs. In contrast, though AHF-Net has a larger number of parameters, there are not many extra operations in the process. It has a lower time cost in large datasets such as the Beijing dataset.

Our algorithm has a smaller number of parameters compared to other algorithms. However, the ICA module needs to sort and reorganize each channel of the feature block. It increases the computations and lowers the time efficiency, so the running time of our algorithm is fast but not the fastest. The remaining models, DMIL, GAFnet, and CRHFF, have time costs proportional to the number of parameters, which is logical and reasonable.

G. Experimental Result With Hohhot Area Dataset, Nanjing Area Dataset, and Shanghai Area Dataset

We conduct the same experiments on the other three datasets, and the experimental results show that our proposed TSMF-Net still has the highest OA, AA, and KAPPA.

For the Hohhot area dataset, the experimental results are shown in Fig. 9 and Table IX. Our proposed algorithm

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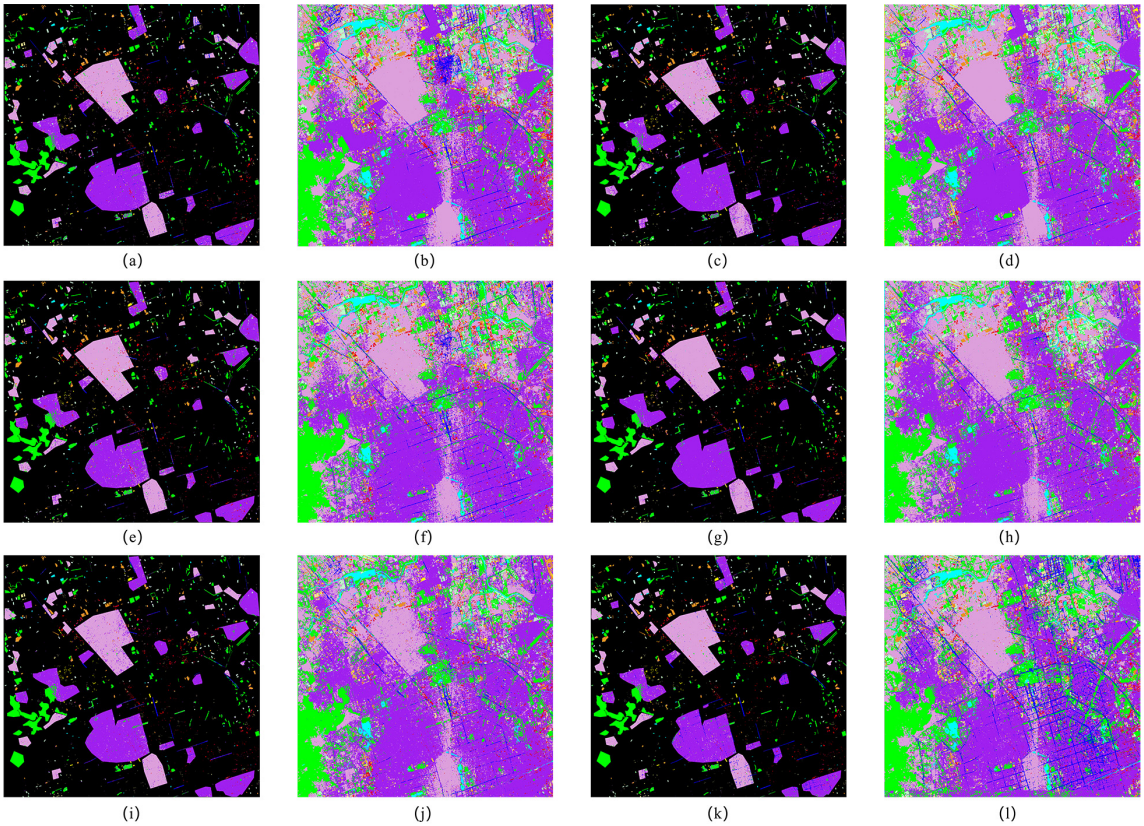

Fig. 8. Classification results of each method on the Beijing dataset: one is the image with ground truth, while the other is the overall image. (a) and (b) ACO-SS + DBFA-Net. (c) and (d) AHF-Net. (e) and (f) GAFnet. (g) and (h) CRHFF. (i) and (j) MFT. (k) and (l) TSMF-Net.

TABLE VIII
QUANTITATIVE CLASSIFICATION RESULTS ON THE BEIJING AREA DATASET (%)

Methods	ACO-SS+DBFA-Net	AHF-Net	GAFnet	CRHFF	MFT	OURS
c_1	78.59±0.19	88.10±0.27	91.81±0.22	89.90±0.16	83.93±0.30	**91.64±0.15**
c_2	70.44±0.28	80.17±0.25	75.22±0.17	88.34±0.23	74.67±0.19	**94.55±0.12**
c_3	70.16±0.12	82.17±0.14	87.93±0.15	91.20±0.12	93.75±0.17	**94.02±0.22**
c_4	82.13±0.17	87.43±0.19	83.77±0.22	88.97±0.21	84.09±0.26	**89.29±0.14**
c_5	76.63±0.25	66.02±0.38	64.45±0.34	87.53±0.27	84.80±0.33	**90.42±0.32**
c_6	85.10±0.10	91.13±0.12	91.78±0.25	90.01±0.29	96.61±0.12	**96.83±0.17**
c_7	88.97±0.22	85.29±0.19	84.77±0.13	93.28±0.24	**95.64±0.19**	92.69±0.15
c_8	78.03±0.24	85.23±0.22	89.98±0.29	94.79±0.12	92.09±0.22	**95.53±0.16**
c_9	76.62±0.15	80.63±0.24	80.36±0.12	**90.06±0.08**	84.85±0.22	89.66±0.19
c_{10}	79.93±0.20	79.67±0.13	73.50±0.22	86.63±0.22	94.17±0.10	**94.45±0.20**
OA	84.21±0.23	85.95±0.20	86.91±0.20	90.13±0.19	93.85±0.20	**94.03±0.15**
AA	77.46±0.21	76.54±0.18	74.35±0.19	90.07±0.22	88.46±0.15	**92.89±0.15**
KAPPA	76.32±0.17	78.65±0.19	79.97±0.15	90.09±0.19	90.70±0.13	**91.14±0.14**
Time(s)	4483.20±373.15	**3002.40±258.32**	9090.84±897.07	4096.74±297.34	3274.70±300.22	3542.14±244.73
Parameters(n)	1.16×10^7	3.94×10^6	4.28×10^7	3.41×10^6	2.90×10^5	1.21×10^6

achieves the highest accuracy in most categories, especially on c6 (Buding2) and c9 (Buidling4). Categories c6 and c9 have rich spatial and spectral information, and our module emphasizes the mutual fusion of the MS and PAN images in the whole process to improve their discrimination ability. ACO-SS + DBFA-Net has the highest accuracy on c5 (Building1) because there are more categories around c5 than other categories, and the adaptive sampling strategy of ACO-SS + DBFA-Net can effectively reduce the confusion of categories.

For the Nanjing area dataset, the experimental results are shown in Fig. 10 and Table X. Overall, the dataset is simple and relatively balanced in terms of categories. However, we found that two pairs of categories, c2 (Road1) and c10 (Road2) and c4 (H-VEG) and c11 (L-VEG), are easily misclassified by analyzing the confusion matrix. By observing the remote sensing images, we found that these categories are widely distributed and have few differences within the category, which is difficult to distinguish. Our proposed TSMF-Net can obtain more balanced spatial and spectral information, and

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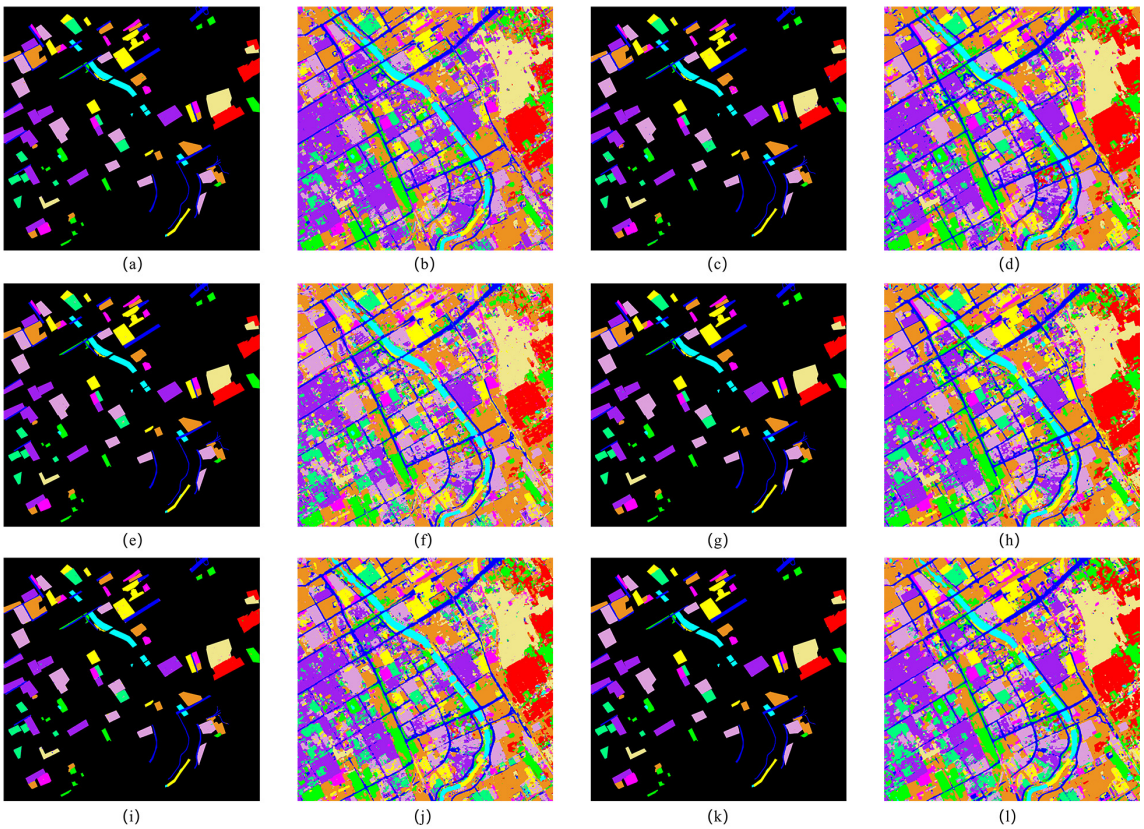

Fig. 9. Classification results of each method on the Hohhot dataset: one is the image with ground truth, while the other is the overall image. (a) and (b) ACO-SS + DBFA-Net. (c) and (d) AHF-Net. (e) and (f) GAFnet. (g) and (h) CRHFF. (i) and (j) MFT. (k) and (l) TSMF-Net.

TABLE IX
QUANTITATIVE CLASSIFICATION RESULTS ON THE HOHHOT AREA DATASET (%)

Methods	ACO-SS+DBFA-Net	AHF-Net	GAFnet	CRHFF	MFT	OURS
c_1	99.21±0.04	99.22±0.18	99.18±0.10	98.56±0.07	99.10±0.12	**99.68±0.05**
c_2	92.24±0.18	94.35±0.34	91.33±0.23	96.98±0.12	97.07±0.19	**97.72±0.24**
c_3	94.04±0.21	96.25±0.32	94.26±0.27	98.31±0.18	97.68±0.17	**99.32±0.21**
c_4	97.60±0.19	96.81±0.29	90.44±0.34	98.47±0.21	97.34±0.31	**99.34±0.29**
c_5	**99.97±0.01**	98.52±0.36	90.96±0.19	97.94±0.08	98.15±0.12	99.23±0.20
c_6	91.14±0.22	97.23±0.18	89.34±0.22	96.99±0.11	98.46±0.23	**99.57±0.17**
c_7	96.11±0.31	98.10±0.27	93.61±0.22	99.03±0.19	97.88±0.21	**99.40±0.15**
c_8	96.42±0.24	97.83±0.31	96.32±0.18	98.80±0.20	99.02±0.19	**99.50±0.23**
c_9	92.60±0.23	96.22±0.35	95.39±0.26	98.26±0.11	97.79±0.28	**99.34±0.30**
c_{10}	92.27±0.19	96.15±0.27	92.10±0.13	97.93±0.16	97.28±0.08	**99.18±0.22**
c_{11}	97.51±0.17	95.69±0.25	90.28±0.15	97.35±0.23	97.58±0.24	**98.49±0.09**
OA	94.80±0.18	96.82±0.19	92.54±0.20	98.24±0.12	97.97±0.19	**99.17±0.15**
AA	95.38±0.24	96.64±0.14	92.09±0.13	98.06±0.08	97.94±0.14	**99.16±0.07**
KAPPA	94.18±0.15	96.27±0.13	91.64±0.19	98.09±0.10	97.73±0.16	**99.07±0.13**
Time(s)	454.21±31.23	412.18±28.84	374.31±29.52	431.79±25.96	322.94±30.44	427.89±29.66

it fully fuses essential and peculiar features. Thus, it can obtain satisfactory results in these difficult categories. Moreover, our method also has good results in other categories, achieving a satisfactory overall performance.

For the Shanghai area dataset, the experimental results are shown in Fig. 11 and Table XI. From the experimental results, we can learn that c1 (Water), c2 (Road), and c8 (Building3) belong to the difficult categories. Among them, c2 (Road) has a wide distribution, complex surroundings, and abundant adjacent categories, which makes it hard to classify. Our algorithm emphasizes fusing spatial and spectral information, and some essential features are overly strengthened in some cases, which leads to confusion between categories and degrades the classification accuracy. However, this situation is relatively rare and does not occur in the other three datasets, so it does not affect the overall advancement of our algorithm. For the difficult categories of buildings such as c8, our algorithm achieves good results with significantly better accuracy than other algorithms. It is because our network can fuse spectral and spatial information mutually, and even if the buildings

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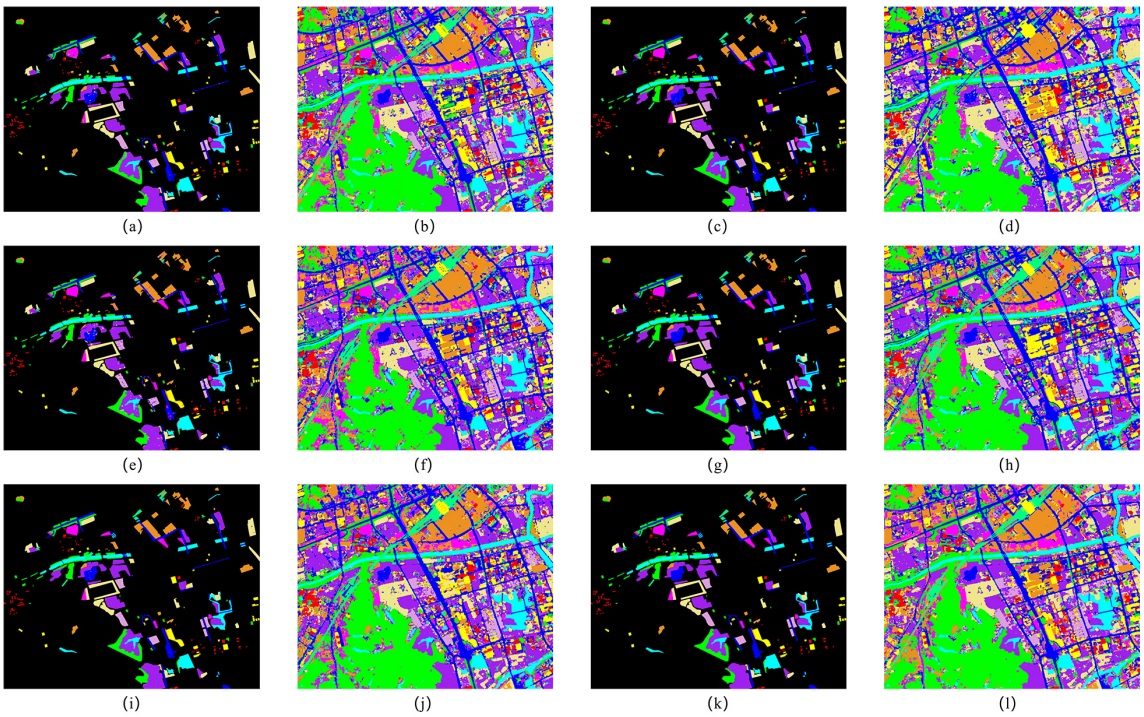

Fig. 10. Classification maps of different methods on the Nanjing area images, each method has two classification maps: one is the image with ground truth, while the other is the overall image. (a) and (b) ACO-SS + DBFA-Net. (c) and (d) AHF-Net. (e) and (f) GAFnet. (g) and (h) CRHFF. (i) and (j) MFT. (k) and (l) TSMF-Net.

TABLE X
QUANTITATIVE CLASSIFICATION RESULTS ON THE NANJING AREA DATASET (%)

Methods	ACO-SS+DBFA-Net	AHF-Net	GAFnet	CRHFF	MFT	OURS
c_1	96.97±0.21	97.02±0.17	93.73±0.31	98.01±0.14	97.03±0.23	**98.48±0.18**
c_2	93.69±0.33	93.51±0.46	88.40±0.62	97.23±0.31	97.53±0.44	**98.06±0.35**
c_3	98.36±0.18	97.26±0.41	94.85±0.25	99.47±0.10	98.72±0.17	**99.69±0.04**
c_4	97.16±0.34	97.55±0.51	93.95±0.37	98.63±0.19	97.84±0.23	**99.11±0.12**
c_5	96.28±0.25	97.26±0.49	92.24±0.33	**99.76±0.05**	98.84±0.17	99.70±0.09
c_6	98.12±0.11	99.45±0.32	93.99±0.41	99.37±0.15	**99.84±0.03**	99.79±0.08
c_7	97.59±0.29	95.89±0.44	95.11±0.35	**99.56±0.08**	98.37±0.14	98.91±0.22
c_8	99.16±0.10	97.98±0.65	96.48±0.29	**99.24±0.07**	98.60±0.17	99.07±0.10
c_9	95.36±0.39	97.19±0.77	82.02±0.81	98.76±0.28	98.67±0.35	**99.67±0.08**
c_{10}	97.02±0.22	97.50±0.67	95.13±0.57	98.19±0.31	97.84±0.34	**99.15±0.29**
c_{11}	95.23±0.44	94.48±0.75	83.27±0.67	98.49±0.21	97.77±0.30	**99.33±0.19**
OA	95.71±0.23	96.13±0.24	92.54±0.31	98.91±0.11	98.13±0.20	**99.02±0.14**
AA	94.99±0.25	96.46±0.22	92.77±0.24	98.79±0.14	98.28±0.15	**99.18±0.13**
KAPPA	95.03±0.17	95.51±0.13	91.37±0.19	98.71±0.09	97.83±0.14	**98.87±0.14**
Time(s)	306.15±29.98	423.41±35.72	382.00±25.36	380.63±30.64	339.95±32.17	391.33±33.58

have similar spatial information, our algorithm can still project the features into a feature space where they can be easily distinguished, thus improving the classification accuracy of these categories. It is worth mentioning that MFT performs well in the building categories, achieving the highest accuracy on c5 (Building1) and c7 (Building2), which is closely related to its powerful modeling capability.

H. Ablation Study

In this section, we analyze the application effects and contribution of different strategies and methods by a combination of them. We take Hohhot dataset as an example, combining the baseline module (Resnet18), ATIHS strategy, and FG-Block for ablation study. Specifically, each combination has the same hyperparameters and experimental setup. The results of the ablation study are shown in Table XII.

As shown in the table, the OA of the model with ATIHS strategy improves by 6.11% (97.61/91.55) from Baseline, while AA and KAPPA also improve accordingly. The stability of the classification also goes up. However, the improvement of the model applying FG-Block alone is not significant because FG-Block requires homogeneity of the feature blocks of the two branches. Without ATIHS for data fusion, the gap between the feature blocks is too large, resulting in a slight improvement. Correspondingly, the accuracy improvement of the model with both application of ATIHS and FG-Block is

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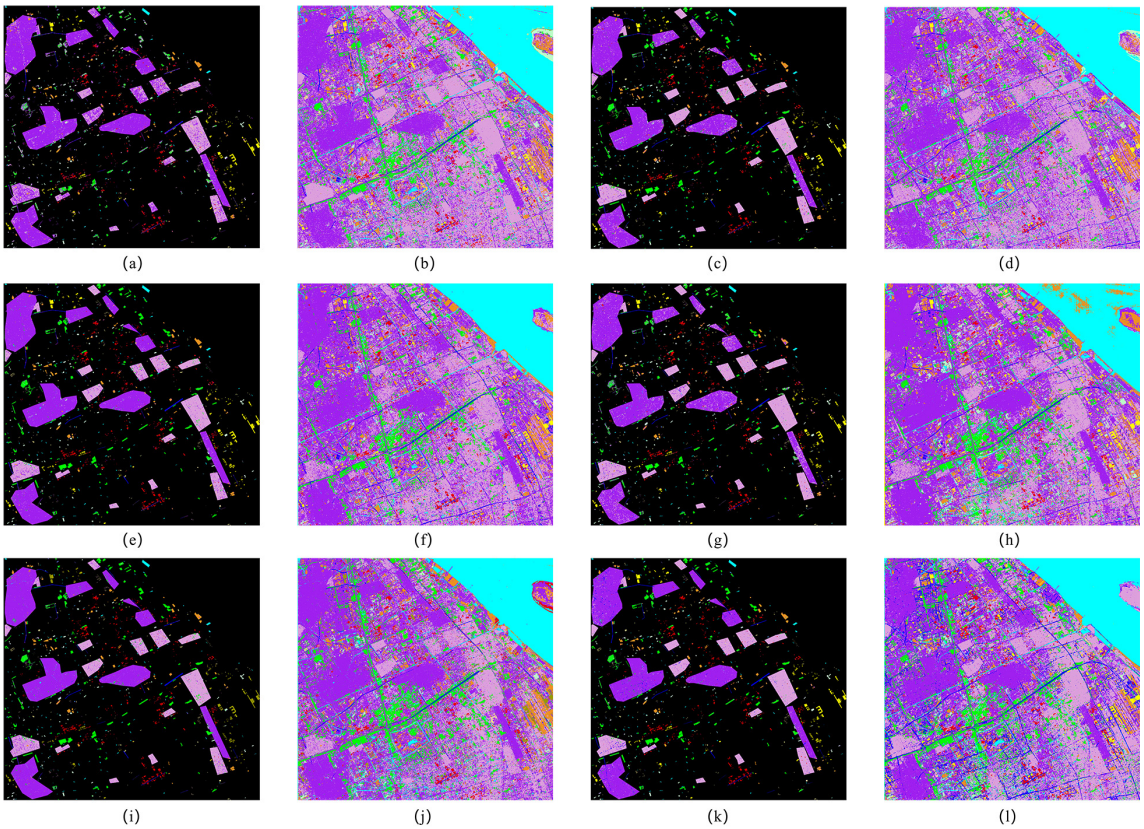

Fig. 11. Classification maps of different methods on the Shanghai area images, each method has two classification maps: one is the image with ground truth, while the other is the overall image. (a) and (b) ACO-SS + DBFA-Net. (c) and (d) AHF-Net. (e) and (f) GAFnet. (g) and (h) CRHFF. (i) and (j) MFT. (k) and (l) TSMF-Net.

TABLE XI

QUANTITATIVE CLASSIFICATION RESULTS ON THE SHANGHAI AREA DATASET (%)

Methods	ACO-SS+DBFA-Net	AHF-Net	GAFnet	CRHFF	MFT	OURS
c_1	88.40±0.20	83.71±0.13	64.35±0.19	**89.02±0.05**	85.63±0.12	87.90±0.13
c_2	**91.47±0.14**	52.90±0.07	49.01±0.17	89.96±0.11	86.26±0.17	86.11±0.20
c_3	90.85±0.15	77.90±0.36	93.75±0.22	92.57±0.24	**96.43±0.11**	93.21±0.09
c_4	86.63±0.33	64.76±0.22	80.79±0.30	89.73±0.17	88.48±0.23	**95.54±0.11**
c_5	84.91±0.29	80.28±0.19	88.20±0.18	91.34±0.22	**95.84±0.02**	95.24±0.12
c_6	84.48±0.18	76.67±0.25	79.04±0.21	87.57±0.15	94.01±0.14	**97.68±0.10**
c_7	85.19±0.19	76.66±0.21	93.21±0.17	91.76±0.11	**96.92±0.19**	91.62±0.07
c_8	76.95±0.23	83.01±0.28	76.74±0.21	86.96±0.20	87.19±0.08	**97.56±0.11**
c_9	92.70±0.25	65.61±0.14	69.35±0.15	92.98±0.15	85.31±0.13	**95.14±0.09**
c_{10}	93.38±0.16	93.61±0.18	92.24±0.17	95.31±0.06	96.08±0.14	**97.42±0.03**
OA	75.48±0.20	78.38±0.23	86.65±0.18	90.04±0.14	93.69±0.16	**93.78±0.08**
AA	83.49±0.19	75.51±0.16	78.67±0.18	90.72±0.15	91.22±0.19	**93.74±0.10**
KAPPA	72.32±0.22	65.56±0.13	77.62±0.19	89.97±0.09	89.58±0.15	**90.04±0.11**
Time(s)	3918.69±336.09	**3100.89±370.23**	3389.78±407.76	3594.32±360.81	4697.48±310.73	3298.26±342.66

TABLE XII

ABLATION EXPERIMENT RESULTS OF EACH COMPONENT ON THE HOHHOT AREA DATASET (%)

Baseline	ATIHS	FG-Block	OA	AA	KAPPA
✓			91.55	90.74	90.52
	✓		97.61	97.33	97.33
		✓	93.98	93.18	93.25
	✓	✓	99.17	99.16	99.07

excellent, which fully exploits the advantages of FG-Block, confirming the previous view.

V. CONCLUSION

In this article, we propose the TSMF-Net for PAN and MS image classification. Experiments on multiple datasets have proved the effectiveness of our proposed algorithms. Our proposed ATIHS strategy has satisfactory fusion results and effectively suppresses the spectral distortion, and the q-Split operation reduces the structural differences and expands the fusion approach. We innovatively introduce the triplet loss and propose a tradeoff fusion module ICA in FG-Block, achieving good performance and broadening the design ideas. However, TSMF-Net is not an end-to-end framework, and the calculation

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amount of the ICA module is a bit large. In the near future, we will write coefficient calculation in Python and efficient channel rearrangement algorithms to better cope with the task of remote sensing image classification.

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